

LAB ASSIGNMENT- 9

Name: MOURIMA SAHA

Roll No: CSE 326

Table: members

member_id	name	plan	city	fee
1	Aarav	Gold	Delhi	5000
2	Meera	Silver	Mumbai	3000
3	Rohan	Gold	Bangalore	5500
4	Sneha	Platinum	Delhi	7000
5	Kunal	Silver	Chennai	3200

Table : sessions

session_id	member_id	trainer	duration	calories
101	1	Raj	60	400
102	2	Simran	45	300
103	3	Raj	70	500
104	4	Aman	80	600
105	5	Raj	50	350

Questions:

Q1. Display all members

Query: db.members.find()

Output:

```
{
  _id: ObjectId('69da79d1d32aa099ed1e2621'),
  member_id: 1,
  name: 'Aarav',
  plan: 'Gold',
  city: 'Delhi',
  fee: 5000
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2622'),
  member_id: 2,
  name: 'Meera',
  plan: 'Silver',
  city: 'Mumbai',
  fee: 3000
},
```

```
{
  _id: ObjectId('69da79d1d32aa099ed1e2623'),
  member_id: 3,
  name: 'Rohan',
  plan: 'Gold',
  city: 'Bangalore',
  fee: 5500
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2624'),
  member_id: 4,
  name: 'Sneha',
  plan: 'Platinum',
  city: 'Delhi',
  fee: 7000
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2625'),
  member_id: 5,
  name: 'Kunal',
  plan: 'Silver',
  city: 'Chennai',
  fee: 3200
}
```

Q2. Show name and plan only

Query: db.members.find({}, {name:1, plan:1, _id:0})

Output:

```
[
  { name: 'Aarav', plan: 'Gold' },
  { name: 'Meera', plan: 'Silver' },
  { name: 'Rohan', plan: 'Gold' },
  { name: 'Sneha', plan: 'Platinum' },
  { name: 'Kunal', plan: 'Silver' }
]
```

Q3. Find members from Delhi

Query: db.members.find({city:"Delhi"})

Output:

```
[
  {
    _id: ObjectId('69da79d1d32aa099ed1e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69da79d1d32aa099ed1e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  }
]
```

Q4. Find sessions with duration > 60

Query: db.sessions.find({duration:{\$gt:60}})

Output:

```
[
  {
    _id: ObjectId('69da79dbd32aa099ed1e2628'),
    session_id: 103,
    member_id: 3,
    trainer: 'Raj',
    duration: 70,
    calories: 500
  },
  {
    _id: ObjectId('69da79dbd32aa099ed1e2629'),
    session_id: 104,
    member_id: 4,
    trainer: 'Aman',
    duration: 80,
    calories: 600
  }
]
```

Q5. Sort members by fee descending

Query: `db.members.find().sort({fee:-1})`

Output:

```
{
  _id: ObjectId('69da79d1d32aa099ed1e2624'),
  member_id: 4,
  name: 'Sneha',
  plan: 'Platinum',
  city: 'Delhi',
  fee: 7000
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2623'),
  member_id: 3,
  name: 'Rohan',
  plan: 'Gold',
  city: 'Bangalore',
  fee: 5500
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2621'),
  member_id: 1,
  name: 'Aarav',
  plan: 'Gold',
  city: 'Delhi',
  fee: 5000
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2625'),
  member_id: 5,
  name: 'Kunal',
  plan: 'Silver',
  city: 'Chennai',
  fee: 3200
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2622'),
  member_id: 2,
  name: 'Meera',
  plan: 'Silver',
  city: 'Mumbai',
  fee: 3000
}
```

Q6. Update fee +500 for Gold members

Query: `db.members.updateMany({plan:"Gold"}, {$inc:{fee:500}})`

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q7. Delete members with fee < 3500

Query: db.members.deleteMany({fee:{<3500}})

Output: { acknowledged: true, deletedCount: 2 }

Q8. Insert a new member

Query: db.members.insertOne({member_id:6, name:"Ankit", plan:"Gold", city:"Kolkata", fee:6000})

Output:{acknowledged: true,

insertedId: ObjectId('69da8066d32aa099ed1e262b')}

Q9. Average fee per plan

Query: db.members.aggregate([{\$group:{_id:"\$plan", avgFee:{\$avg:"\$fee"}}}])

Output:

```
[
  { _id: 'Gold', avgFee: 5833.333333333333 },
  { _id: 'Platinum', avgFee: 7000 }
]
```

Q10. Total calories burned per trainer

Query: db.sessions.aggregate([{\$group:{_id:"\$trainer", totalCalories:{\$sum:"\$calories"}}}])

Output:

```
[
  { _id: 'Simran', totalCalories: 300 },
  { _id: 'Raj', totalCalories: 1250 },
  { _id: 'Aman', totalCalories: 600 }
]
```

Q11. Maximum fee per city

Query: db.members.aggregate([{\$group:{_id:"\$city", maxFee:{\$max:"\$fee"}}}])

Output:

```
[
  { _id: 'Bangalore', maxFee: 6000 },
  { _id: 'Delhi', maxFee: 7000 },
  { _id: 'Kolkata', maxFee: 6000 }
]
```

Q12. Count members per plan

Query: db.members.aggregate([{\$group:{_id:"\$plan", count:{\$sum:1}}}]

Output:

```
[ { _id: 'Gold', count: 3 }, { _id: 'Platinum', count: 1 } ]
```

Q13. Trainers with avg calories > 400

Query: db.sessions.aggregate([{\$group:{_id:"\$trainer", avgCalories:{\$avg:"\$calories"}},
{\$match:{avgCalories:{\$gt:400}}}]

Output:

```
[  
  { _id: 'Aman', avgCalories: 600 },  
  { _id: 'Raj', avgCalories: 416.6666666666667 }  
]
```

Q14. Total duration per member_id

Query: db.sessions.aggregate([{\$group:{_id:"\$member_id", totalDuration:{\$sum:"\$duration"}}}]

Output:

```
[  
  { _id: 4, totalDuration: 80 },  
  { _id: 3, totalDuration: 70 },  
  { _id: 1, totalDuration: 60 },  
  { _id: 2, totalDuration: 45 },  
  { _id: 5, totalDuration: 50 }  
]
```

Q15. Join members with sessions

Query: db.members.aggregate([{\$lookup:{from:"sessions", localField:"member_id",
foreignField:"member_id", as:"session_details"}}}]

Output:

```

{
  _id: ObjectId('69da79d1d32aa099ed1e2621'),
  member_id: 1,
  name: 'Aarav',
  plan: 'Gold',
  city: 'Delhi',
  fee: 5500,
  session_details: [
    {
      _id: ObjectId('69da79dbd32aa099ed1e2626'),
      session_id: 101,
      member_id: 1,
      trainer: 'Raj',
      duration: 60,
      calories: 400
    }
  ]
},
{
  _id: ObjectId('69da79d1d32aa099ed1e2623'),
  member_id: 3,
  name: 'Rohan',
  plan: 'Gold',
  city: 'Bangalore',
  fee: 6000,
  session_details: [
    {
      _id: ObjectId('69da79dbd32aa099ed1e2628'),
      session_id: 103,
      member_id: 3,
      trainer: 'Raj',
      duration: 70,
      calories: 500
    }
  ]
},

```

```

},
{
  _id: ObjectId('69da79d1d32aa099ed1e2624'),
  member_id: 4,
  name: 'Sneha',
  plan: 'Platinum',
  city: 'Delhi',
  fee: 7000,
  session_details: [
    {
      _id: ObjectId('69da79dbd32aa099ed1e2629'),
      session_id: 104,
      member_id: 4,
      trainer: 'Aman',
      duration: 80,
      calories: 600
    }
  ]
},
{
  _id: ObjectId('69da8066d32aa099ed1e262b'),
  member_id: 6,
  name: 'Ankit',
  plan: 'Gold',
  city: 'Kolkata',
  fee: 6000,
  session_details: []
}

```

Q16. Show members who have sessions

Query: `db.members.aggregate([{$lookup:{from:"sessions", localField:"member_id", foreignField:"member_id", as:"session"}}, {$match:{"session.0":{"$exists:true}}}]`)

Output:

```
[
  {
    _id: ObjectId('69da79d1d32aa099ed1e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    session: [
      {
        _id: ObjectId('69da79dbd32aa099ed1e2626'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69da79d1d32aa099ed1e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    session: [
      {
        _id: ObjectId('69da79dbd32aa099ed1e2628'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  }
]
```

```
{
  _id: ObjectId('69da79d1d32aa099ed1e2624'),
  member_id: 4,
  name: 'Sneha',
  plan: 'Platinum',
  city: 'Delhi',
  fee: 7000,
  session: [
    {
      _id: ObjectId('69da79dbd32aa099ed1e2629'),
      session_id: 104,
      member_id: 4,
      trainer: 'Aman',
      duration: 80,
      calories: 600
    }
  ]
}
```


Q17. Create index on city

Query: `db.members.createIndex({city:1})`

Output: city_1

Q19. Create compound index on plan and fee

Query: `db.members.createIndex({plan:1, fee:-1})`

Output: plan_1_fee_-1

Q20. Display 2nd and 3rd highest fee members

Query: `db.members.find().sort({fee:-1}).skip(1).limit(2)`

Output:

```
[
  {
    _id: ObjectId('69da8066d32aa099ed1e262b'),
    member_id: 6,
    name: 'Ankit',
    plan: 'Gold',
    city: 'Kolkata',
    fee: 6000
  },
  {
    _id: ObjectId('69da79d1d32aa099ed1e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000
  }
]
```

Compass output: members collection

```
▶ _id: ObjectId('69da79d1d32aa099ed1e2621')
  member_id: 1
  name: "Aarav"
  plan: "Gold"
  city: "Delhi"
  fee: 5500
```

```
_id: ObjectId('69da79d1d32aa099ed1e2623')
member_id: 3
name: "Rohan"
plan: "Gold"
city: "Bangalore"
fee: 6000
```

```
_id: ObjectId('69da79d1d32aa099ed1e2624')
member_id: 4
name: "Sneha"
plan: "Platinum"
city: "Delhi"
fee: 7000
```



```
_id: ObjectId('69da8066d32aa099ed1e262b')
member_id: 6
name: "Ankit"
plan: "Gold"
city: "Kolkata"
fee: 6000
```

Compass output: sessions collection

```
_id: ObjectId('69da79dbd32aa099ed1e2626')
session_id: 101
member_id: 1
trainer: "Raj"
duration: 60
calories: 400
```



```
_id: ObjectId('69da79dbd32aa099ed1e2627')
session_id: 102
member_id: 2
trainer: "Simran"
duration: 45
calories: 300
```

102

```
_id: ObjectId('69da79dbd32aa099ed1e2628')
session_id: 103
member_id: 3
trainer: "Raj"
duration: 70
calories: 500
```



```
_id: ObjectId('69da79dbd32aa099ed1e2629')
session_id: 104
member_id: 4
trainer: "Aman"
duration: 80
calories: 600
```

```
_id: ObjectId('69da79dbd32aa099ed1e262a')
session_id: 105
member_id: 5
trainer: "Raj"
duration: 50
calories: 350
```